Day 10: Data Visualization with Matplotlib

AI and Machine Learning Course

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What is Matplotlib?

- A Python library that lets you **draw graphs and charts**
- Helps you **see** your data instead of just reading numbers
- Works with NumPy and Pandas

Getting Started

- Install: `pip install matplotlib`
- Import: `import matplotlib.pyplot as plt`

How to Make a Plot — 5 Steps

1. Prepare your data (lists, arrays, DataFrame)
2. Pick the right chart type
3. Call the plot function (e.g. `plt.plot()`)
4. Add title and labels
5. Show it with `plt.show()`

Line Plot — Show Trends

- Use when data changes **over time**
- Example: temperature each month, stock price each day
- Function: `plt.plot(x, y)`

Option	**What it does**
`color='red'`	Change line color
`marker='o'`	Add dots at each point
`linestyle='--'`	Make the line dashed
`label='...'`	Name for the legend

Bar Chart — Compare Groups

- Use when comparing **different categories**
- Example: marks in different subjects, sales by product
- `plt.bar(names, values)` — vertical bars
- `plt.barh(names, values)` — horizontal bars

Histogram — See How Data is Spread

- Use to see the **shape** of your data
- Example: how many students scored 60–70, 70–80, etc.
- `plt.hist(data, bins=10)`
- `bins` = number of groups to split the data into

Histogram

- Numbers (continuous)
- Bars touch each other

Bar Chart

- Categories (names)
- Bars have gaps

Scatter Plot & Pie Chart

Scatter Plot

- Shows if two things are **related**
- Example: study hours vs marks
- `plt.scatter(x, y)`

Pie Chart

- Shows **parts of a whole**
- Example: market share
- `plt.pie(sizes, labels=names)`

Subplots — Multiple Charts in One Figure

- Put several charts side by side for comparison
- `fig, axes = plt.subplots(rows, cols)`
- Use `axes[0, 0]`, `axes[0, 1]`, etc. to access each chart
- `plt.tight_layout()` — fixes overlapping

Making Plots Look Good

What to add	How
Title	`plt.title("My Title")`
X-axis name	`plt.xlabel("...")`
Y-axis name	`plt.ylabel("...")`
Legend	`plt.legend()`
Grid lines	`plt.grid(True)`
Save as image	`plt.savefig('plot.png')`

Tip

Always add a title and axis labels so anyone can understand your chart!

Which Chart Should I Use?

Chart	Function	Use When...
Line	`plt.plot()`	Data changes over time
Bar	`plt.bar()`	Comparing groups
Histogram	`plt.hist()`	Seeing data spread
Scatter	`plt.scatter()`	Checking if two things are related
Pie	`plt.pie()`	Showing parts of a whole